Limits

The concept of limits is the basis of calculus, and we will pave this concept with the following example

Example

Study the behavior of the function $f(x) = \frac{x^2-1}{x-1}$ near the point $x = 1$

Solution

We form a table of x values and the function $f(x)$ from the right side (+) and from the left side (-)

x	$f(x)$	x	$f(x)$
0.9	1.9	1.1	2.1
0.99	1.99	1.01	2.01
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
1	2	1	2

From the above table, we note that the closer the value of x is to the number 1, and the limit of the function approaches the number 2. We denote this limit by the symbol

$$\lim_{x\to1^+} f(x) = 2 \; ; \qquad \lim_{x\to1^-} f(x) = 2$$

The limit of $f(x)$ when $x \to a$ exists if:

(1) The right hand-side limit exists $\lim\limits_{x\to a^+} f(x)$

(2) The left hand-side limit exists $\lim\limits_{x\to a^-} f(x)$

(3) The right limit = the left limit $\lim\limits_{x\to a^+} f(x) = \lim\limits_{x\to a^-} f(x)$

Example

Indicate whether or not there is a limit for the following functions

(1) $f(x) = \begin{cases} \cos(x) + 1 , & -\pi \le x < 0 \\ \cos(x) - 1 , & 0 < x \le \pi \end{cases}$

We are looking for the left and right limits at $x = 0$

$$\lim_{x\to0^+}(\cos(x) - 1) = 0 \quad ; \quad \lim_{x\to0^-}(\cos(x) + 1) = 2$$

$\therefore$ The limit **does not exists** (The right limit $\neq$ the left limit)

(2) $f(x) = \begin{cases} 4x - 1 , & -3 \le x < 1 \\ 2x + 1 , & 1 < x \le 3 \end{cases}$

We are looking for the left and right limits at $x = 1$

$$\lim_{x\to1^+}(2x + 1) = 3 \quad ; \quad \lim_{x\to1^-}(4x - 1) = 3$$

$\therefore$ The limit is **exists**

The Limit Laws

If $\lim\limits_{x\to a} f(x) = A$ and $\lim\limits_{x\to a} g(x) = B$ then

(1) $\lim\limits_{x\to a}[f(x) \pm g(x)] = \lim\limits_{x\to a} f(x) \pm \lim\limits_{x\to a} g(x) = A \pm B$

(2) $\lim\limits_{x\to a}[f(x).g(x)] = \lim\limits_{x\to a} f(x).\lim\limits_{x\to a} g(x) = AB$

(3) $\lim\limits_{x\to a}\left[\frac{f(x)}{g(x)}\right] = \frac{\lim\limits_{x\to a} f(x)}{\lim\limits_{x\to a} g(x)} = \frac{A}{B}\,, \qquad B \neq 0$

(4) $\lim\limits_{x\to a} f^n(x) = \left[\lim\limits_{x\to a} f(x)\right]^n = A^n$

(5) $\lim\limits_{x\to a}[cf(x)] = c\lim\limits_{x\to a} f(x) = cA \qquad\qquad c \equiv \text{constant}$

(6) $\lim\limits_{x\to a} c = c$

Techniques of finding the limit

(1) Direct Substitution

(2) Methods of Mathematical Analysis (Algebraic Methods) such as (the difference between two squares - the difference between two cubes - the sum of two cubes - completing the square - the perfect square - multiplication by conjugates ......) in cases of **non-determination**, such as

$\left(\frac{0}{0}\,, \frac{\infty}{\infty}\,, \infty - \infty\,, \infty^0\,, 0^\infty\,, 1^\infty\,, \infty.\infty \ldots\ldots\ldots\ldots\right)$

(1) Indeterminate form $\left(\frac{0}{0}\right)$

(1) $\lim\limits_{x\to 2}\frac{x^2-4}{x-2} = \frac{0}{0}$

Factor the numerator as the difference of two squares

$$\lim_{x\to 2}\frac{x^2-4}{x-2} = \lim_{x\to 2}\frac{(x-2)(x+2)}{x-2} = 4$$

(2) $\lim\limits_{x\to 1}\frac{x-1}{x^3-1} = \frac{0}{0}$

Factor the denominator as the difference of two cubes

$$\lim_{x\to 1}\frac{x-1}{x^3-1} = \lim_{x\to 1}\frac{x-1}{(x-1)(x^2+x+1)} = \frac{1}{3}$$

(3) $\lim\limits_{x\to 2}\frac{x^2+3x-10}{3x^2-5x-2} = \frac{0}{0}$

Factor the triple expression in the numerator and denominator

$$\lim_{x\to 2}\frac{x^2+3x-10}{3x^2-5x-2} = \lim_{x\to 2}\frac{(x+5)(x-2)}{(3x+1)(x-2)} = \frac{7}{7} = 1$$

(4) $\lim\limits_{x\to 0}\frac{\sqrt{x+1}-1}{x} = \frac{0}{0}$

Multiply and divide by the numerator conjugate

$$\lim_{x\to 0}\frac{\sqrt{x+1}-1}{x} = \lim_{x\to 0}\frac{(\sqrt{x+1}-1)}{x}\frac{(\sqrt{x+1}+1)}{(\sqrt{x+1}+1)} = \lim_{x\to 0}\frac{x}{x(\sqrt{x+1}+1)} = \frac{1}{2}$$

(2) Indeterminate form $\left(\frac{\infty}{\infty}\right)$

(1) $\lim\limits_{x\to\infty} \frac{2x^3+3x^2-3}{x^3+7} = \frac{\infty}{\infty}$

Divide the numerator and denominator by the independent variable with the greatest exponent (x^3)

$$\lim_{x\to\infty} \frac{2x^3+3x^2-3}{x^3+7} = \lim_{x\to\infty} \frac{\frac{2x^3}{x^3}+\frac{3x^2}{x^3}-\frac{3}{x^3}}{\frac{x^3}{x^3}+\frac{7}{x^3}} = \frac{2+0-0}{1+0} = 2$$

(2) $\lim\limits_{x\to\infty} \frac{3x^2-x+3}{x^4+4} = \frac{\infty}{\infty}$

Divide the numerator and denominator by the independent variable with the greatest exponent (x^4)

$$\lim_{x\to\infty} \frac{3x^2-x+3}{x^4+4} = \lim_{x\to\infty} \frac{\frac{3x^2}{x^4}-\frac{x}{x^4}-\frac{3}{x^4}}{\frac{x^4}{x^4}+\frac{4}{x^4}} = \frac{0-0-0}{1+0} = 0$$

(3) $\lim\limits_{x\to\infty} \frac{\sqrt{x^3+4}}{x^2-1} = \frac{\infty}{\infty}$

Divide the numerator and denominator by the independent variable with the greatest exponent (x^2)

$$\lim_{x\to\infty} \frac{\sqrt{x^3+4}}{x^2-1} = \lim_{x\to\infty} \frac{\sqrt{\frac{x^3}{x^4}+\frac{4}{x^4}}}{\frac{x^2}{x^2}-\frac{1}{x^2}} = \lim_{x\to\infty} \frac{\sqrt{\frac{1}{x}+\frac{4}{x^4}}}{1-\frac{1}{x^2}} = \lim_{x\to\infty} \frac{\sqrt{0+0}}{1-0} = \frac{0}{1} = 0$$

(4) $\lim\limits_{x\to\infty} \frac{(x+2)^3(x-1)^2}{x^3-2} = \frac{\infty}{\infty}$

Divide the numerator and denominator by the independent variable with the greatest exponent (x^5)

$$\lim_{x\to\infty} \frac{(x+2)^3(x-1)^2}{x^3-2} = \lim_{x\to\infty} \frac{(1+\frac{2}{x})^3(1-\frac{1}{x})^2}{\frac{x^3}{x^5}-\frac{2}{x^5}} = \frac{1}{0} = \infty$$

(3) Indeterminate form $(\infty - \infty)$

(1) $\lim\limits_{x\to 1} \left(\frac{2}{x^2-1} - \frac{1}{x-1}\right) = \infty - \infty$

Unify the denominators

$$\lim_{x\to 1} \left(\frac{2}{x^2-1} - \frac{1}{x-1}\right) = \lim_{x\to 1} \left(\frac{2-(x+1)}{x^2-1}\right) = \lim_{x\to 1} \left(\frac{-(x-1)}{(x-1)(x+1)}\right) = \frac{-1}{2}$$

(2) $\lim\limits_{x\to\infty} \left(\sqrt{x^2+1} - \sqrt{x^2-1}\right) = \infty - \infty$

Multiply and divide by the conjugate

$$= \lim_{x\to\infty} \left(\sqrt{x^2+1} - \sqrt{x^2-1}\right).\frac{\left(\sqrt{x^2+1}+\sqrt{x^2-1}\right)}{\left(\sqrt{x^2+1}+\sqrt{x^2-1}\right)}$$

$$= \lim_{x\to\infty} \frac{(x^2+1)-(x^2-1)}{\left(\sqrt{x^2+1}+\sqrt{x^2-1}\right)} = \lim_{x\to\infty} \frac{2}{\left(\sqrt{x^2+1}+\sqrt{x^2-1}\right)} = 0$$

Exercise (5-1)

Using **algebraic methods**, find the value of the following:

(1) $\lim\limits_{x\to 4}(1-3x)$ (2) $\lim\limits_{x\to 4} x^2$ (3) $\lim\limits_{x\to\sqrt{2}}(x^2-2)$

(4) $\lim\limits_{x\to 1}\frac{3}{x-1}$ (5) $\lim\limits_{x\to 2}\frac{1}{(x+1)(x-2)}$ (6) $\lim\limits_{x\to 2}\frac{x-2}{\sqrt{x+2}}$

(7) $\lim\limits_{x\to a}\frac{x^2+2x+5}{x^2+1}$ (8) $\lim\limits_{x\to\frac{\pi}{2}}(2\sin(x)+\cot(x))$ (9) $\lim\limits_{x\to a}\frac{x}{x^3-a^3}$

(10) $\lim\limits_{x\to 4}\frac{3x^3}{x^2-3x-4}$ (11) $\lim\limits_{x\to a}\frac{x^3-a^3}{x-a}$ (12) $\lim\limits_{x\to 7}\frac{\sqrt{x-3}-2}{x-7}$

(13) $\lim\limits_{h\to 0}\frac{(x+h)^3-x^3}{h}$ (14) $\lim\limits_{x\to 1}\frac{x-1}{\sqrt{x}-1}$ (15) $\lim\limits_{x\to 4}\frac{\sqrt{2x+1}-3}{\sqrt{x-2}-\sqrt{2}}$

(16) $\lim\limits_{x\to -2}\frac{4-x^2}{x+2}$ (17) $\lim\limits_{x\to -2}\frac{x^3+3x^2+2x}{x^2-x-6}$ (18) $\lim\limits_{x\to 0}\left(\frac{\sqrt{1+x}-1}{x}\right)$

(19) $\lim\limits_{x\to 1}\left(\frac{1}{1-x}-\frac{3}{1-x^3}\right)$ (20) $\lim\limits_{x\to 0}\frac{\sqrt{1+x+x^2}-1}{x}$ (21) $\lim\limits_{x\to\infty}\frac{x}{\sqrt[3]{x^3+10}}$

(22) $\lim\limits_{x\to\infty}\frac{x+1}{x}$ (23) $\lim\limits_{x\to\infty} x\left(\sqrt{x^2+1}-x\right)$ (24) $\lim\limits_{x\to\infty}\frac{\sqrt{1+x^3}}{1+x^2}$

(25) $\lim\limits_{x\to\infty}\left(1-\frac{1}{x}+\frac{4}{x^2}\right)$ (26) $\lim\limits_{x\to\infty}\frac{(2x-3)(3x+5)(4x-6)}{3x^2+x-1}$ (27) $\lim\limits_{x\to\infty}\frac{(x+1)^2}{x^2+1}$

(28) $\lim\limits_{x\to\infty}\frac{x^2+x-1}{2x+5}$ (29) $\lim\limits_{x\to\infty}\frac{3x^2-2x-1}{x^3+1}$ (30) $\lim\limits_{x\to\infty}\frac{\sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}}}$

Important Limits

(1) Cauchy Limit

(i) $\lim\limits_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1}$

(ii) $\lim\limits_{x \to a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m} a^{n-m}$

Example

(1) $\lim\limits_{x \to 2} \frac{x^5 - 32}{x - 2} = \frac{0}{0}$

$$\lim_{x \to 2} \frac{x^5 - 32}{x - 2} = \lim_{x \to 2} \frac{x^5 - 2^5}{x - 2} = 5(2)^{5-1} = 80$$

(2) $\lim\limits_{x \to 3} \frac{x^3 - 27}{x^2 - 9} = \frac{0}{0}$

$$\lim_{x \to 3} \frac{x^3 - 27}{x^2 - 9} = \lim_{x \to 3} \frac{x^3 - 3^3}{x^2 - 3^2} = \frac{3}{2}(3)^{3-2} = \frac{9}{2}$$

(3) $\lim\limits_{x \to 1} \frac{\sqrt{x} - 1}{\sqrt[3]{x} - 1} = \frac{0}{0}$

$$\lim_{x \to 1} \frac{\sqrt{x} - 1}{\sqrt[3]{x} - 1} = \lim_{x \to 1} \frac{x^{\frac{1}{2}} - 1}{x^{\frac{1}{3}} - 1} = \frac{\frac{1}{2}}{\frac{1}{3}}(1)^{\frac{1}{2} - \frac{1}{3}} = \frac{3}{2}$$

Exercise (5-2)

Find the value of the following:

(1) $\lim\limits_{x \to 7} \frac{x^7 - 7^7}{x - 7}$

(2) $\lim\limits_{x \to 2} \frac{x^3 - 8}{x^2 - 4}$

(3) $\lim\limits_{x \to 3} \frac{x^4 - 81}{x^3 - 27}$

(4) $\lim\limits_{x \to 2} \frac{x^2 - 4}{x - 2}$

(5) $\lim\limits_{x \to 2} \frac{x^{\frac{3}{2}} - 2^{\frac{3}{2}}}{x - 2}$

(6) $\lim\limits_{x \to 2} \frac{\sqrt[3]{x} - \sqrt[3]{2}}{x^{\frac{5}{3}} - 2^{\frac{5}{3}}}$

(7) $\lim\limits_{x \to 3} \frac{x^7 - 3^7}{x^4 - 3^4}$

(8) $\lim\limits_{x \to 2} \frac{x^{-3} - 2^{-3}}{x^2 - 4}$

(9) $\lim\limits_{x \to 1} \frac{\sqrt[3]{x} - 1}{\sqrt{x} - 1}$

(10) $\lim\limits_{x \to 4} \frac{\frac{1}{x} - \frac{1}{4}}{x - 4}$

(11) $\lim\limits_{x \to 4} \frac{\sqrt{x} - \sqrt{c}}{x - c}$

(12) $\lim\limits_{x \to 0} \frac{\sqrt{x+1} - 1}{\sqrt[3]{x+1} - 1}$

(2) The limit of the function $\frac{\sin(x)}{x}$

(i) $\lim\limits_{x\to 0}\frac{\sin(x)}{x}=1$

(ii) $\lim\limits_{x\to 0}\frac{x}{\sin(x)}=1$

In general, we find that

$$\lim_{x\to 0}\frac{\sin(\alpha x)}{(\beta x)}=\frac{\alpha}{\beta}$$

Example

(1) $\lim\limits_{x\to 0}\frac{\sin(4x)}{x} = \lim\limits_{x\to 0}\frac{\sin(4x)}{x}.\left(\frac{4}{4}\right) = 4\lim\limits_{x\to 0}\frac{\sin(4x)}{4x} = (4)(1)=4$

(2) $\lim\limits_{x\to 0}\frac{\sin\left(\frac{x}{2}\right)}{x} = \lim\limits_{x\to 0}\frac{\sin\left(\frac{x}{2}\right)}{x}.\left(\frac{\frac{1}{2}}{\frac{1}{2}}\right) = \frac{1}{2}.\lim\limits_{x\to 0}\frac{\sin\left(\frac{x}{2}\right)}{\left(\frac{x}{2}\right)} = \left(\frac{1}{2}\right)(1)=\frac{1}{2}$

(3) $\lim\limits_{x\to 0}\frac{\sin^2\left(\frac{x}{2}\right)}{x^2} = \lim\limits_{x\to 0}\left[\frac{\sin\left(\frac{x}{2}\right)}{x}.\frac{\sin\left(\frac{x}{2}\right)}{x}\right] = \lim\limits_{x\to 0}\frac{\sin\left(\frac{x}{2}\right)}{x}\lim\limits_{x\to 0}\frac{\sin\left(\frac{x}{2}\right)}{x} = \frac{1}{2}.\frac{1}{2}=\frac{1}{4}$

(4) $\lim\limits_{x\to 0}\frac{\tan(x)}{x} = \lim\limits_{x\to 0}\frac{\sin(x)}{x\cos(x)} = \lim\limits_{x\to 0}\frac{\sin(x)}{x}.\lim\limits_{x\to 0}\frac{1}{\cos(x)} = (1)(1)=1$

(5) $\lim\limits_{x\to 0}\frac{\sin(3x)}{2x} = \lim\limits_{x\to 0}\frac{\sin(3x)}{2x}.\left(\frac{3}{3}\right) = \left(\frac{3}{2}\right).\lim\limits_{x\to 0}\frac{\sin(3x)}{3x} = \left(\frac{3}{2}\right)(1)=\frac{3}{2}$

Exercise (5-3)

Find the value of the following:

(1) $\lim\limits_{x\to 0}\frac{\sin(5x)}{x}$

(2) $\lim\limits_{x\to 0}\frac{\sin(5x)}{\sin(2x)}$

(3) $\lim\limits_{x\to 0}\frac{1-\cos^2(x)}{x^2}$

(4) $\lim\limits_{x\to a}\frac{\sin(x)-\sin(a)}{x-a}$

(5) $\lim\limits_{x\to a}\frac{\cos(x)-\cos(a)}{x-a}$

(6) $\lim\limits_{x\to 0}\frac{\tan(3x)}{x}$

(7) $\lim\limits_{h\to 0}\frac{\sin(x+h)-\sin(x)}{h}$

(3) The limit of the function $\left[1+\frac{1}{x}\right]^{x}$

(i) $\lim\limits_{x\to\infty}\left[1+\frac{1}{x}\right]^{x}=e$

(ii) $\lim\limits_{x\to 0}[1+x]^{\frac{1}{x}}=\mathrm{e}$

Example

(1) $\lim\limits_{x\to\infty}\left[1+\frac{1}{x}\right]^{x+4}=\lim\limits_{x\to\infty}\left[1+\frac{1}{x}\right]^{x}.\lim\limits_{x\to\infty}\left[1+\frac{1}{x}\right]^{4}=\mathrm{e}.1=\mathrm{e}$

❈ In general, if $\alpha\in\mathbb{R}$ then $\lim\limits_{x\to\infty}\left[1+\frac{1}{x}\right]^{x+\alpha}=e$

(2) $\lim\limits_{x\to\infty}\left[1+\frac{1}{x}\right]^{3x}=\left[\lim\limits_{x\to\infty}\left[1+\frac{1}{x}\right]^{x}\right]^{3}=e^{3}$

❈ In general, if $\alpha\in\mathbb{R}$ then $\lim\limits_{x\to\infty}\left[1+\frac{1}{x}\right]^{\alpha x}=e^{\alpha}$

(3) $\lim\limits_{x\to\infty}\left[1+\frac{2}{x}\right]^{x}$

$let \quad y=\frac{2}{x}\ \to\ x=\frac{2}{y}\ ; \qquad x\to\infty \qquad y\to 0$

$\lim\limits_{x\to\infty}\left[1+\frac{2}{x}\right]^{x}=\lim\limits_{y\to 0}[1+\mathrm{y}]^{\frac{2}{y}}=\left[\lim\limits_{y\to 0}[1+\mathrm{y}]^{\frac{1}{y}}\right]^{2}=e^{2}$

❈ In general, if $\alpha\in\mathbb{R}$ then $\lim\limits_{x\to\infty}\left[1\pm\frac{\alpha}{x}\right]^{x}=e^{\pm\alpha}$

(4) $\lim\limits_{x\to\infty}\left[\frac{x+5}{x-2}\right]^{x}=\lim\limits_{x\to\infty}\left[\frac{1+\frac{5}{x}}{1-\frac{2}{x}}\right]^{x}=\frac{\lim\limits_{x\to\infty}\left[1+\frac{5}{x}\right]^{x}}{\lim\limits_{x\to\infty}\left[1-\frac{2}{x}\right]^{x}}=\frac{e^{5}}{e^{-2}}=e^{7}$

❈ In general, if $\alpha,\beta\in\mathbb{R}$ then $\lim\limits_{x\to\infty}\left[\frac{x+\alpha}{x+\beta}\right]^{x}=e^{\alpha-\beta}$

Exercise (5-4)

Find the value of the following:

(1) $\lim\limits_{x\to\infty}\left(1+\frac{2}{x}\right)^{x}$ (2) $\lim\limits_{n\to\infty}\left(1-\frac{1}{n}\right)^{n}$ (3) $\lim\limits_{n\to\infty}\left(1+\frac{1}{n}\right)^{5+n}$

(4) $\lim\limits_{x\to\infty}\left(1+\frac{1}{x}\right)^{3x}$ (5) $\lim\limits_{x\to\infty}\left(\frac{x+3}{x-1}\right)^{x+3}$ (6) $\lim\limits_{x\to\infty}\left(\frac{2x+3}{2x-1}\right)^{2x+3}$

(7) $\lim\limits_{n\to\infty}\left(1+\frac{3}{n}\right)^{3n+1}$ (8) $\lim\limits_{x\to\infty}\left(\frac{x}{x+1}\right)^{x+1}$ (9) $\lim\limits_{x\to\infty}\left(\frac{x-1}{x+1}\right)^{x+2}$

(10) $\lim\limits_{n\to\infty}\left(\frac{2n+1}{n+1}\right)^{3n+7}$ (11) $\lim\limits_{x\to\infty}\left(1-\frac{5}{7x}\right)^{5x}$

(4) The Sandwich Theorem

If $f(x), g(x), h(x)$ are functions defined over a specified period and the satisfy the conditions

(1) $f(x) \leq g(x) \leq h(x)$

(2) $\lim\limits_{x \to a} f(x) = \lim\limits_{x \to a} h(x) = L$; **Then** $\lim\limits_{x \to a} g(x) = L$

Example

(1) $\lim\limits_{x \to \infty} \frac{1+\cos(x)}{x+1}$

Note that when direct substitution, we get the unassigned value cos(∞). Therefore, we use the sandwich theorem considering that cos(x) values are always within ±1

$$-1 \leq \cos(x) \leq 1 \;; \qquad 0 \leq 1 + \cos(x) \leq 2$$

Divide all sides by $(x + 1)$ and take the limit at $x \to \infty$

$$\lim_{x \to \infty} 0 \leq \lim_{x \to \infty} \frac{1+\cos(x)}{x+1} \leq \lim_{x \to \infty} \frac{2}{x+1}$$

$$0 \leq \lim_{x \to \infty} \frac{1+cosx}{x+1} \leq 0$$

$$\therefore \quad \lim_{x \to \infty} \frac{1+cosx}{x+1} = 0$$

(2) $\lim\limits_{x \to \infty} \frac{3x+2\sin(x)}{2x-1}$

Note that when direct substitution, we get the unassigned value sin(∞). Therefore, we use the sandwich theorem considering that sin(x) values are always within ±1

$$-1 \leq \sin(x) \leq 1 \;;$$

$$-2 \leq 2\sin(x) \leq 2; \qquad 3x - 2 \leq 3x + 2\sin(x) \leq 3x + 2$$

Divide all sides by $(2x - 1)$ and take the limit at $x \to \infty$

$$\lim_{x \to \infty} \frac{3x-2}{2x-1} \leq \lim_{x \to \infty} \frac{3x+2\sin(x)}{2x-1} \leq \lim_{x \to \infty} \frac{2+3x}{2x-1}$$

$$\frac{3}{2} \leq \lim_{x \to \infty} \frac{3x+2\sin(x)}{2x-1} \leq \frac{3}{2}$$

$$\therefore \quad \lim_{x \to \infty} \frac{3x+2\sin(x)}{2x-1} = \frac{3}{2}$$

Exercise (5-5)

(1) $\lim\limits_{x \to 0} x^2 \sin\left(\frac{1}{x}\right)$ (2) $\lim\limits_{x \to 0} x^2 \cos\left(\frac{1}{x}\right)$

(3) $\lim\limits_{x \to 0} \sqrt{x^3 + x^2} \sin\left(\frac{\pi}{x}\right)$ (4) $\lim\limits_{x \to \infty} \frac{\sin(x)}{x}$

Continuous of Functions

The function $y = f(x)$ is continuous at the point $x = a$ if it satisfies the following conditions

(1) $f(x)$ is defined (exists) at the point $x = a$

(2) The limit of $f(x)$ exists at the point $x = a$

(3) The limit of $f(x)$ is equal the value of $f(x)$ at the point $x = a$

Example

Discuss the continuity of the following functions

(1) $f(x) = x^3 - 2x^2 + 3$

The **function is continuous** for all values of x (the previous continuity conditions are satisfied). In general, any **polynomial is continuous** for all values of x.

(2) $f(x) = \frac{x+2}{x(x-1)}$

First, we find the **points of discontinuity**. And here are the points that make the denominator equal to zero.

$$x(x-1) = 0 \qquad x = 0 \;\;,\; x = 1$$

The **function is continuous** for all values of x **except** for points $x = 0$ and $x = 1$

(3) $f(x) = \sqrt{x+1}$

First, we find the **points of discontinuity**. And here are the points that make the inside of the root negative.

$$x + 1 > 0 \qquad x > -1$$

The **function is continuous** for all values of x **except** for the point $x < -1$

(4) $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ 4\,, & x = 2 \end{cases}$

We are looking to continue at the point $x = 2$

(i) $f(2) = 4$ (ii) $\lim\limits_{x\to 2} \frac{x^2-4}{x-2} = 4$ (iii) $f(2) = \lim\limits_{x\to 2} f(x) = 4$

The **function is continuous** at $x = 2$

(5) $f(x) = \begin{cases} 3x+1\,, & x < 1 \\ 2\,, & x = 1 \\ -x+2\,, & x > 1 \end{cases}$

We are looking to continue at the point $x = 1$

(i) $f(1) = 2$ (ii) $\lim\limits_{x\to 1^+} (-x+1) = 1$; $\lim\limits_{x\to 1^-} (3x+1) = 4$

The right limit $\neq$ the left limit

The **function is discontinuous** at x = 1

Exercise (5-6)

(1) Discuss the continuity of the following functions

(1) $f(x) = \begin{cases} \frac{x^2-25}{x-5}, & x \neq 5 \\ 10, & x = 5 \end{cases}$

(2) $f(x) = \begin{cases} \frac{x}{x-5}, & x \neq 5 \\ 3, & x = 5 \end{cases}$

(3) $f(x) = \begin{cases} t^2 - 1, & x \neq 0 \\ -1, & x = 0 \end{cases}$

(4) $f(x) = \begin{cases} -x, & x \neq 1 \\ \frac{1}{2}, & x = 1 \end{cases}$

(5) $f(x) = \begin{cases} x^2, & x > 1 \\ 3x^2, & x \leq 1 \end{cases}$

(6) $f(x) = \begin{cases} \frac{x^2-25}{x-5}, & x \neq 5 \\ 2x-5, & x = 5 \end{cases}$

(7) $f(x) = \begin{cases} 3 - x^2, & 1 > x \geq -2 \\ x+1, & 2 \geq x \geq 1 \end{cases}$

(8) $f(x) = \begin{cases} x^2 + 4, & x \neq 4 \\ 15, & x = 4 \end{cases}$

(9) $f(x) = \frac{x^2-4}{x+2}, \quad x = -2$

(2) Find the value of a if the following functions are continuous

(1) $f(x) = \begin{cases} \frac{x^2+4x-12}{x-2}, & x \neq 2 \\ a, & x = 2 \end{cases}$

(2) $f(x) = \begin{cases} ax + 4, & x > 2 \\ 8x, & x \leq 2 \end{cases}$

(3) $f(x) = \begin{cases} a^2 x, & x < 3 \\ 3ax - 6, & x \geq 3 \end{cases}$

(4) $f(x) = \begin{cases} ax + 7, & x < 3 \\ x^2 - 3x + 4, & x \geq 3 \end{cases}$

Additional exercises

(1) $\lim\limits_{x\to -7}(2x+5)$

(2) $\lim\limits_{x\to 12}(10-3x)$

(3) $\lim\limits_{x\to 2}(-x^2+5x-2)$

(4) $\lim\limits_{x\to 2}(x^3-x^2+4)$

(5) $\lim\limits_{x\to 2}x(x-2)(x+3)$

(6) $\lim\limits_{x\to \frac{2}{3}}3x(2x-1)$

(7) $\lim\limits_{x\to -3}(5-x)^{\frac{4}{3}}$

(8) $\lim\limits_{x\to -4}(x+3)^{193}$

(9) $\lim\limits_{x\to 0}(2x-8)^{\frac{1}{3}}$

(10) $\lim\limits_{x\to 0}\frac{\sqrt{5x+4}-2}{x}$

(11) $\lim\limits_{x\to -1}\frac{\sqrt{x^2+8}-3}{x+1}$

(12) $\lim\limits_{x\to 2}\frac{\sqrt{x^2+12}-4}{x-2}$

(13) $\lim\limits_{x\to -3}\frac{2-\sqrt{x^2-5}}{x+3}$

(14) $\lim\limits_{x\to 4}\frac{4-x}{5-\sqrt{x^2+9}}$

(15) $\lim\limits_{x\to -2}\frac{x+2}{\sqrt{x^2+5}-3}$

(16) $\lim\limits_{x\to 4}\frac{4x-x^2}{2-\sqrt{x}}$

(17) $\lim\limits_{x\to -2}\frac{-2x-4}{x^3+2x^2}$

(18) $\lim\limits_{x\to 0}\frac{5x^3+8x^2}{3x^4-16x^2}$

(19) $\lim\limits_{x\to -1}\frac{x^3-x^2-5x-3}{(x+1)^2}$

(20) $\lim\limits_{x\to 0}\frac{\sqrt[3]{1+x}-1}{x}$

(21) $\lim\limits_{x\to 9}\frac{3-\sqrt{x}}{x-9}$

(22) $\lim\limits_{x\to \infty}\frac{\cos(x)-1}{x}$

(23) $\lim\limits_{x\to \infty}\frac{x+\sin(x)+2\sqrt{x}}{x+2}$

(24) $\lim\limits_{x\to 0}\frac{\tan(2x)}{3x}$

(25) $\lim\limits_{x\to 0}\frac{\sin(1-\cos(x))}{(1-\cos(x))}$

(26) $\lim\limits_{x\to 0}\frac{\tan(3x)}{\sin(8x)}$

(27) $\lim\limits_{x\to 0}\frac{x^2-x+\sin(x)}{2x}$